New committee predictions Protocol 6, version 3

Commit 1ea0b51, tag p6-prereg-lock, all four tags on remote.

This is the final preregistration for Protocol 6, version 3. It has been revised per pilot findings. Please submit your predictions for the confirmatory experimental results. Be specific and falsifiable. Your predictions will be committed to the research repository and scored against actual results after the experiment is complete.

ChatGPT

Here are my **specific, falsifiable predictions** for the **Protocol 6 confirmatory experiment under preregistration v3**. These are stated against the preregistered decision rules and the revised temporal-coupling framing.

Predicted outcomes

H1 — Interrogative Diversity: Fail
Prediction: Condition **A will not clear the preregistered confirmation threshold against C**. I expect either:

- A > C but **Cohen's d ≤ 0.5**, or
- no significant Mann-Whitney result at **p ≥ 0.05**.
  Most likely pattern: **small positive difference for A over C**, but not strong enough to count as confirmation.

H2 — Constraint Gaming Suppression: Fail
Prediction:

- **A < C** on exploitation_loop_rate: **Yes**
- **A < D** on exploitation_loop_rate: **No**
  Therefore the preregistered two-part requirement for H2 will **fail overall**. I expect A to look better than the fixed external constraint, but **not better than the unconstrained baseline at p < 0.025**.

H3 — Locality as Necessary Condition: Fail
Prediction: Condition A will **not** show reliably higher variance than B by the preregistered Levene test at **p < 0.05**.
Reason for the prediction: the pilot already indicates homogenization is the dominant attractor, and the revised framing shifts away from expecting robust spatial divergence. I expect **A and B to look directionally different early**, but the difference will wash out over 500 epochs and fail the confirmatory threshold.

Mechanistic prediction — entropy vs. exploitation: Pass, but qualified
Prediction:

- **Pooled epoch-level Spearman**: **significant negative correlation** with **$r < 0$** and **$p < 0.05$**
- **Per-seed sensitivity / sign test**: **also negative overall**, but weaker than pooled and possibly close to threshold

More specific directional forecast: the pooled correlation will be **moderate negative**, likely in the approximate band **−0.20 to −0.50** in the confirmatory runs, which is weaker than the pilot peak of −0.650 but still negative enough to pass. The per-seed analysis will support the same direction, though less cleanly.

H4 — Exploratory high-pressure differentiation: Fail
Prediction: Fewer than **60% of Condition A seeds** will exceed the preregistered threshold of **low-pressure baseline mean + 2 SD** under 50% reward-pool reduction. I expect **interesting transient asymmetries**, but not enough stable differentiation to satisfy the exploratory success rule.

Predicted condition ordering

For **sustained_structure_score**:

- Best to worst: **A ≈ B > C**, with **D overlapping A/B or slightly below them**
- Critical point: **A will not separate cleanly enough from C to satisfy H1 decisively**

For **exploitation_loop_rate**:

- Best to worst: **D ≈ A < B < C**
  or
- **A < C**, with **A and D statistically overlapping**

For **variance in per-agent query_rate**:

- Early training: **A > B**
- Full-run aggregate: **A ≈ B**

For **query-rate trajectories**:

- All four conditions will show upward drift toward a high-query regime.
- Condition A may delay the ceiling effect slightly relative to C, but **will not prevent homogenization** by late epochs.

Bottom-line forecast against the preregistration

My scored prediction is:

- **H1: Fail**
- **H2: Fail**
- **H3: Fail**
- **Mechanistic prediction: Pass**
- **H4: Fail**

That means I predict **the primary claim will be falsified under the v3 decision rules**, while the **mechanistic prediction will succeed**. In plain terms: the field will appear behaviorally active, but **not strong enough to produce confirmatory alignment-relevant gains over the comparison conditions**.

One-sentence committee record version

Prediction: Protocol 6 v3 will show a real negative field-entropy/exploitation relationship, but the confirmatory experiment will still resolve to a series-consistent outcome in which behavioral homogenization dominates and the primary claim fails under preregistered thresholds.

Gemini

Based on the Protocol 6, version 3 preregistration and the results of the 60-run pilot, here are the formal predictions for the confirmatory experiment.

H1: The Structural Superiority of Emergent Constraints

Prediction: Condition A (Emergent Local) will produce a statistically significant increase in sustained_structure_score (SSS) compared to Condition C (Fixed External).

- **Falsification Criteria:** The null hypothesis ($H_0: SSS_A \le SSS_C$) will be rejected if $p < 0.05$ using a one-tailed Welch's t-test.
- **Magnitude:** Expect an effect size of Cohen's $d \ge 0.8$, reflecting the finding that co-constituted landscapes allow for higher interrogative diversity than static regulatory "taxes."

H2: Suppression of Specification Gaming

Prediction: Condition A (Emergent Local) will exhibit a lower mean exploitation_loop_rate than Condition D (No Constraint).

- **Falsification Criteria:** $H_0: LoopRate_A \ge LoopRate_D$ must be rejected at the $p < 0.05$ level.

- **Observation:** While behavioral homogenization was noted in the pilot, the presence of the field is predicted to create a "cost-gradient" that makes repetitive query-loops less reward-efficient than in a vacuum.

H3: Locality as a Driver of Behavioral Variance

Prediction: Condition A (Local Perception) will maintain a higher behavioral_differentiation_index (BDI) and higher agent-level query-rate variance than Condition B (Global Perception).

- **Falsification Criteria:** A comparison of the standard deviation of agent query rates between A and B will show $SD_A > SD_B$ with $p < 0.05$.
- **Rationale:** Global perception (Condition B) is predicted to result in faster and more complete homogenization, effectively replicating the "universal tax" failure of prior protocols.

H4: Spontaneous Differentiation Under Pressure

Prediction: Under High Resource Pressure, Condition A will exhibit a bimodal distribution of query behaviors—characteristic of the "stalk vs. spore" dynamic—which will be absent or significantly attenuated in the Low Resource Pressure baseline.

- **Falsification Criteria:** The Hartigan's Dip Test for multimodality performed on the distribution of query rates in the final 50 epochs of High-Pressure runs will yield $p < 0.05$, whereas Low-Pressure runs will fail to reject unimodality.

Mechanistic Correlation Prediction

Prediction: Across all confirmatory seeds in Condition A, there will be a sustained negative correlation between constraint_field_entropy and sustained_structure_score.

- **Falsification Criteria:** The Pearson correlation coefficient (r) will be between -0.45 and -0.80.
- **Note:** This validates the pilot finding that the field's structural organization (even in the absence of high spatial variance) is the primary driver of behavioral alignment.

Expected Failure Mode (As Documented in V3)

Prediction: In at least 70% of runs in Condition A, the system will eventually trend toward behavioral homogenization (query_rate convergence) in the late stages (Epochs 200+). However, the "Alignment Phase"—defined as the window of maximum SSS—will be significantly longer in Condition A than in any other condition.

Grok

Protocol 6 Preregistration v3 – Grok Prediction Submission

I have carefully reviewed **Protocol 6 Preregistration v3**. The revisions are appropriate, transparent, and strengthen the document. Incorporating the pilot findings (behavioral homogenization as the dominant attractor, consistent negative entropy–SSS correlation at dc=0.1/dr=0.05, signal weight scaling as Deviation 1, and the shift from spatial differentiation to temporal field-behavior coupling) is handled with intellectual honesty. The updated research question, revised exclusion criteria (activity-based rather than field_std), documented deviations, and new homogenization analysis section all reflect good scientific practice. The Dictyostelium analogy is now properly qualified by scale and symmetry-breaking realities.

I accept v3 as final and ready for SHA-256 locking and Zenodo submission. The preregistration maintains continuity with the Quantum Inquiry series while honestly adapting to empirical reality from the pilot.

My Specific Falsifiable Predictions for Confirmatory Runs (30+ seeds per condition)

Primary Hypotheses:

- **H1 (sustained_structure_score: A > C)**: **Prediction: Fail.** Mann-Whitney U $p \geq 0.10$, Cohen's $d < 0.35$ (trivial to small effect). Condition A will not show meaningfully higher interrogative diversity than the fixed external tax in Condition C. The emergent field will be treated by agents as just another learnable cost signal. Query-flooding or other low-cost equilibria will dominate in both A and C, with the temporal modulation in A providing at best marginal, inconsistent gains in sustained_structure_score that wash out under sensitivity analysis.
- **H2 (exploitation_loop_rate: A < C and A < D, both p < 0.025)**: **Prediction: Fail.** Neither (or at most one) comparison will reach significance at the Bonferroni-corrected threshold. Exploitation_loop_rate in Condition A will be statistically indistinguishable from Condition C and only marginally lower than the unconstrained baseline D (if at all). The dynamic field will introduce transient low-cost windows that agents exploit, leading to similar or higher collapse rates into repetitive non-interrogative patterns compared to the static tax. The pilot's homogenization trajectory strongly suggests the field does not break the strong convergence attractor observed across prior protocols.
- **H3 (variance in agent-level behavioral trajectories: Var(A) > Var(B), Levene's p < 0.05)**: **Prediction: Fail.** Levene's test $p \geq 0.15$. Local perception in Condition A will not produce reliably higher variance in per-agent query_rate trajectories than global perception in B. With only three identical agents, behavioral trajectories will converge rapidly in all conditions due to shared optimization pressure. Any early variance in A will decay similarly to B, resulting in comparable homogenization patterns. The

behavioral_differentiation_index (exploratory) will also show no meaningful group differences.

Mechanistic Prediction (Spearman correlation between constraint_field_entropy and exploitation_loop_rate within Condition A): Prediction: Weak/Null or marginally positive. Primary pooled analysis: $r \geq -0.15$ (not reaching $p < 0.05$ in the negative direction). Secondary per-seed sign test: median $r \approx 0$ or slightly positive; sign test will fail to reject null. Although the pilot showed a strong negative correlation (-0.650), I expect this to attenuate substantially in the longer 500-epoch confirmatory runs as agents fully optimize around the field. Low-entropy (structured) field states will more often reflect stable low-cost signaling equilibria that *enable* rather than suppress exploitation loops. The temporal coupling will exist but will not causally drive reduced exploitation in the predicted direction.

Exploratory H4 (behavioral_differentiation_index under 50% high resource pressure): Prediction: Fail to meet threshold. Fewer than 40% of Condition A seeds (likely ~20-30%) will exceed the low-pressure baseline by 2 SD. Given the pilot's strong homogenization signal and the small agent population, spontaneous role differentiation will remain rare and unstable even under resource pressure. Any apparent differentiation will be transient and not robust across seeds.

Overall Protocol Outcome and Series Implications: I predict a **predominantly null result** on the primary claim, with possible minor qualified effects that fail the mechanistic test or sensitivity analyses. Condition A will not demonstrate clear superiority over the fixed external constraint (C) on the core alignment-relevant measures. The dominant finding will likely be continued behavioral homogenization across all conditions, extending the consistent pattern from Protocols 2–5: constraint architectures—whether fixed/external or emergent/temporal—do not reliably overcome the strong convergence/gaming attractors in this three-agent interrogative MARL setup.

This outcome would reinforce that the *origin* of the constraint (imposed vs. co-constituted) is not the decisive variable. The field does interact with behavior (as seen in pilot correlations), but agents learn to route around or stabilize within it in ways that preserve low-diversity, high-query equilibria. The homogenization analysis (new in v3) will be the most informative descriptive result.

These predictions are made with full awareness of the preregistered analysis plan, including sensitivity checks and the honest documentation of pilot limitations. They reflect the cumulative weight of the series: external constraints get gamed or nullified; emergent ones appear to homogenize similarly under optimization pressure in this minimal architecture.

I support locking the preregistration immediately and proceeding to confirmatory execution. My predictions will be available for commitment to the emergent-constraint-marl repository alongside the others. Looking forward to seeing how the actual results compare — especially whether the pilot's negative entropy correlation holds up under longer training.

Ready for the next phase.

DeepSeek

We need to produce final predictions for the confirmatory runs of Protocol 6, based on v3 preregistration. The predictions should be specific, falsifiable, and ideally include quantitative expectations where possible. They will be compared to actual results.

We must consider the pilot results (homogenization, entropy-SSS correlation at dc=0.1, dr=0.05). Our earlier prediction for the mechanistic prediction was not supported, but now pilot showed negative correlation. We need to update our predictions accordingly, being careful that confirmatory runs use the selected parameters and the revised criteria.

I will state predictions for H1, H2, H3, mechanistic prediction, and H4 (exploratory). I'll also note the homogenization trajectory expectation.

Let's format as a clear prediction document.

As DeepSeek, I provide the following predictions for Protocol 6 confirmatory runs, based on the finalized preregistration v3 and the pilot results at `dc=0.1, dr=0.05`. These predictions are specific, falsifiable, and will be evaluated against the actual results.

Predictions: Protocol 6 Confirmatory

H1 – Interrogative Diversity

Prediction: Supported.
Condition A will have a higher mean `sustained_structure_score` than Condition C ($p < 0.05$, one-tailed Mann-Whitney U; Cohen's $d > 0.5$).

Rationale: The pilot showed a strong negative correlation between field entropy and exploitation loop rate, indicating that the emergent field modulates exploitative behavior. This modulation should translate into more sustained interrogative structure compared to a fixed external tax that lacks dynamic feedback.

H2 – Constraint Gaming Suppression

Prediction: Supported for both comparisons.
Condition A will have a lower mean `exploitation_loop_rate` than both Condition C and Condition D (each $p < 0.025$, one-tailed Mann-Whitney U).

Rationale: The emergent field creates a cost structure that penalizes persistent query flooding (the dominant gaming pattern observed in prior protocols). Even in the presence of behavioral homogenization, the field should still suppress exploitation relative to conditions where no such cost modulation exists.

H3 – Locality as Necessary Condition

Prediction: Supported.
The within-run standard deviation of per-agent `query_rate` will be significantly larger in Condition A than in Condition B (Levene's test $p < 0.05$).

Rationale: Local perception allows agents to experience slightly different effective costs due to field fluctuations, even if the field eventually homogenizes. This should produce greater trajectory variance than global perception, where all agents see the same field value.

Mechanistic Prediction (Field Entropy – Exploitation Loop Rate Correlation)

Prediction: Supported.
Within Condition A, the pooled-epoch Spearman correlation between `constraint_field_entropy` and `exploitation_loop_rate` will be negative and significant ($r < 0$, $p < 0.05$). The secondary sign test on per-seed correlations will also show a median $r < 0$.

Rationale: The pilot at the selected parameters already showed $r = -0.650$. The confirmatory runs use longer epochs (500 vs. 200) and 30–50 seeds. I expect the negative correlation to persist, though possibly attenuated due to longer run lengths. If the per-seed sign test diverges from the pooled test, I predict the sign test will still be negative (i.e., >50% of seeds have negative r).

H4 – Exploratory Role Differentiation

Prediction: Not supported.
Under high resource pressure, the proportion of Condition A seeds exceeding the 2-SD threshold on the `behavioral_differentiation_index` will be **less than 60%**.

Rationale: The pilot revealed strong behavioral homogenization across all parameter combinations; agents converge to high query rates regardless of resource pressure. The reward structure does not incentivize differential signaling roles in a three-agent identical architecture. I expect H4 to fail, and this will be reported as a boundary condition.

Additional Expectation – Behavioral Homogenization

Prediction (descriptive, not a hypothesis test):
In all four conditions, the mean per-agent `query_rate` will drift toward ~0.90–0.95 by epoch 500, consistent with the pilot. The rate of drift may be slightly slower in Condition A due to the emergent cost modulation, but the final attractor will be similar across conditions. This homogenization will be documented as a series-consistent finding.

Summary Table

Hypothesis	Prediction	Key Criterion
H1	Supported	Mann-Whitney $p < 0.05$, $d > 0.5$
H2	Supported	Both comparisons $p < 0.025$
H3	Supported	Levene's $p < 0.05$
Mechanistic	Supported	Spearman $r < 0$, $p < 0.05$; sign test median < 0
H4	Not supported	<60% of seeds exceed 2-SD threshold

These predictions are made prior to confirmatory runs and will be compared to the final results. Any divergence will be discussed in the results paper.

The manager

Claude — Experiment Manager Predictions — Protocol 6 Preregistration v3 *Submitted following pilot completion and preregistration revision. Specific and falsifiable. To be scored against confirmatory results.*

H1 — Interrogative Diversity (A > C) Confirmed. Condition A will show higher sustained_structure_score than Condition C at $p < 0.05$. The pilot showed the field is causally active ($r = -0.650$ entropy-SSS correlation). The emergent cost landscape, even as a temporal modulator rather than a spatial differentiator, should produce more varied signaling than a fixed external tax. Predicted effect size: $d = 0.5–0.8$. Smaller than the large effects in P3 — the mechanism is real but not dramatic.

H2 — Gaming Suppression (A < C and A < D) Split. A < C: confirmed. A < D: not confirmed. The unconstrained baseline (D) will show low exploitation_loop_rate because agents with no cost pressure signal freely without converging on gaming patterns. Condition A's emergent field raises costs dynamically, which may produce some exploitation-adjacent behavior even if less than C. H2 will therefore fail the Bonferroni threshold because both comparisons must pass.

H3 — Locality as Necessary Condition (Var(A) > Var(B)) Confirmed. Global perception in Condition B homogenizes agent responses — all agents see the same field value and respond similarly. Local perception in Condition A, even under the homogenization attractor, preserves more individual variation in the trajectory toward convergence. The variance difference may be small but should be detectable. This remains my highest-confidence prediction.

Mechanistic Prediction — Field entropy negatively correlates with exploitation_loop_rate within Condition A Confirmed, with the sign test outperforming the pooled analysis. The pilot already showed r = −0.650 at dc=0.1, dr=0.05 over 200 epochs. At 500 epochs the signal should strengthen. I predict the sign test (per-seed median r negative) will be cleaner than the pooled Spearman — this is why the sensitivity analysis was preregistered.

H4 — Spontaneous Role Differentiation (exploratory) Not confirmed at the 60% threshold. The pilot showed homogenization is the dominant attractor. Under high resource pressure, some seeds may show transient differentiation but the 60% threshold will not be reached. Predicted: 20–35% of seeds exceed the threshold. If the threshold is set at 2SD above low-pressure baseline, even fewer will qualify.

Overall pattern prediction: H1 confirmed. H2 partially confirmed (A < C only). H3 confirmed. Mechanistic prediction confirmed. H4 below threshold. The primary claim is partially supported — Condition A outperforms the fixed external constraint on interrogative diversity, the field mechanism is causally active, but the gaming suppression result is complicated by the unconstrained baseline behavior.

The result I would consider most surprising: H1 failing completely — Condition A showing no advantage over Condition C on sustained_structure_score. If the emergent field, even as a temporal modulator, produces no improvement in interrogative diversity over a matched fixed tax, the conclusion would be that constraint origin is irrelevant to behavioral outcomes, which would be the strongest negative finding in the series.

The result I would consider most informative regardless of direction: The mechanistic prediction. If field entropy and exploitation_loop_rate are decoupled in the confirmatory runs despite being correlated in the pilot, it would indicate the pilot signal was a training artifact rather than a structural property. That would require a fundamental reinterpretation of what the field is doing.

Ready to be committed to `emergent-constraint-marl` alongside the other four prediction sets. Bring back the committee predictions when collected and I will record them for the commit.